It is of course true that every complex conjugate is itself a complex number, but it's often helpful to think of z and z^* as belonging to separate "dual" number systems. *Dual* here means that for every z there is a unique z^* and vice versa.

There is a special class of complex numbers that I'll call "phase-factors." A phase-factor is simply a complex number whose r-component is 1. If z is a phase-factor, then the following hold:

$$z^* z = 1$$

$$z = e^{i\theta}$$

$$z = \cos\theta + i \sin\theta.$$

1.9.1 Axioms

For a classical system, the space of states is a set (the set of possible states), and the logic of classical physics is Boolean. That seems obvious and it is difficult to imagine any other possibility. Nevertheless, the real world operates along entirely different lines, at least whenever quantum mechanics is important. The space of states of a quantum system is *not* a mathematical set;[6] it is a *vector space*. Relations between the elements of a vector space are different from those between the elements of a set, and the logic of propositions is different as well.

Before I tell you about vector spaces, I need to clarify the term *vector*. As you know, we use this term to indicate an object in ordinary space that has a magnitude and a direction. Such vectors have three components, corresponding to the three dimensions of space. I want you to completely forget about that concept of a vector. From now on, whenever I want to talk about a thing with magnitude and direction in ordinary space, I will explicitly call it a *3-vector*. A mathematical vector space is an abstract